FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2019

PAPER – 2

TEST DATE: 05-05-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

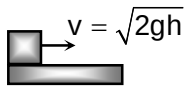
1. C, D

Sol. From initial to final instant

Conservation of momentum gives :

$mv = 2mv'$

$\Rightarrow v' = \frac{\sqrt{2gh}}{2}$



(i)



(f)

Work energy theorem on block + Plank:

$W_f = \frac{1}{2}(2m)\left(\frac{2gh}{4}\right) - \frac{1}{2}m(2gh) = -\frac{mgh}{2}$

Work energy theorem on block:

$W_f = \frac{1}{2}(2m)\left(\frac{2gh}{4}\right) - \frac{1}{2}m(2gh) = -\frac{3}{4}mgh$

$\therefore \left(W_f\right)_{plank} = -\frac{mgh}{2} - \left(-\frac{3}{4}mgh\right) = +\frac{mgh}{4}$

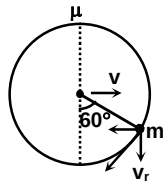
2. C

Sol. Conservation of momentum give:

$Mv = m(v_r \cos 60° - v)$

$\frac{mv_r \cos 60°}{(m+M)} = v$ …(i)

Energy conservation



$$\frac{3}{2}mgR = \frac{1}{2}Mv^2 + \frac{1}{2}m\left[(v_r\cos 60^\circ - v)^2 + (v_r \sin 60^\circ)^2\right] \quad \ldots(ii)$$

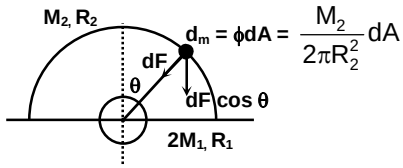
Solve (i) and (ii) for v

3. B, C, D

Sol. Based on blackbody radiation.

4. A, B, C

Sol. $F_N = \int dF\cos\theta = \frac{G(2M_1)}{R_2^2}\frac{M_2}{2\pi R_2^2}\int dA\cos\theta = \frac{GM_1M_2}{R_2^2}$



$\therefore$ The force extended by half shell (M_1, R_1) will be $\frac{GM_1M_2}{2R_2^2}$

So, A, B, C are Correct.

5. A, B, D

Sol. Based on magnetism

6. B, C, D

Sol. From the concept of power in A.C. circuit.

7. C

Sol. Equations are :

$f_1 - f_2 = Ma_C$ …(i)

$(f_1 + f_2)R = \frac{MR^2}{2}\alpha$ …(ii)

$2F - f_1 = Ma_1$ …(iii)

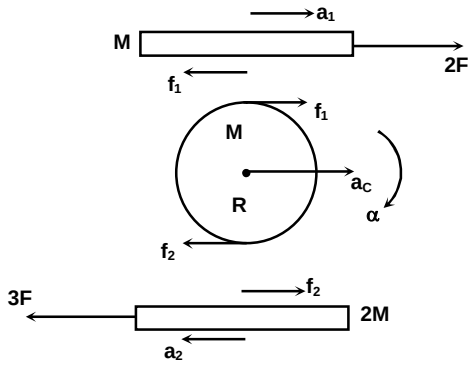
$3F - f_2 = 2Ma_2$ …(iv)

Constraints are

Lower point: $\alpha R - a_C = a_2$ …(v)

Upper point : $\alpha R + a_C = a_1$ …(vi)

Solve for required quantities.



8. B

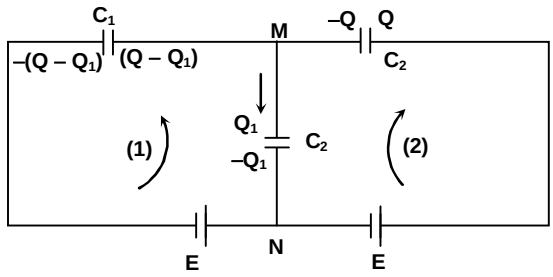
Sol. KVL in L_1 and L_2 gives

$E = \left(\frac{Q - Q_1}{C_1}\right) - \frac{Q_1}{C_2}$ …(i)

$E = \frac{Q}{C_2} - \frac{Q_1}{C_2}$ …(ii)

Solving above

$V_{MN} = \left|E\left(\frac{C_2 - C_1}{2C_2 + C_1}\right)\right|$

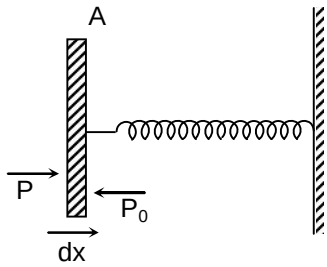


9. C

Sol. Draw FBD and use LOM to solve

10. D

Sol. From Work energy theorem,



$W_{gas} - \int P_0 A dx - \int kx dx = 0$

$\therefore\ W_{gas} = P_0\Delta V + \frac{1}{2}kx^2 = 90\,J$, where x = 0.1 m

Using $\Delta Q = \Delta U + W_{gas}$

$= nC_v\Delta T + W_{gas}$

$= \frac{3}{2}\left[PV - P_0V_0\right] + W_{gas}$

$= \frac{3}{2}\left[PV - P_0V_0\right] + W_{gas} = 2010 + 90 = 2100J$

Calculate the other values similarly

SECTION – D

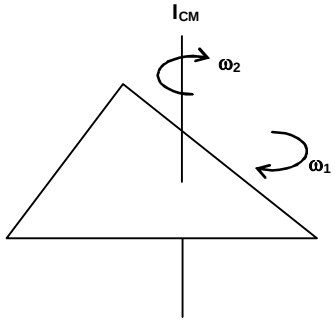
11. 00001.00

Sol. Use constraint that length of thread is constant always.

12. 00000.01

Sol. $\omega_1 \rightarrow$ angular velocity of lamina about I_{CM}.

$\omega_2 \rightarrow$ angular velocity of axis I_{CM}.



$\therefore\ \frac{\vec{dL}}{dt} = \vec{\omega}_2 \times \vec{L}$

$\Rightarrow \left|\frac{\vec{dL}}{dt}\right| = \omega_1\omega_2 I_{CM}$

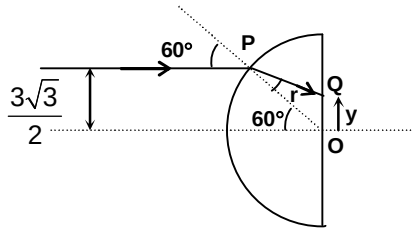
13. 00000.50

Sol. Using $\Delta\ell = \ell\,\alpha\Delta T$

For no displacement, compare the stress on both side of section.

14. 00000.02

Sol. At P, $\sin r = \frac{\sqrt{3}}{2\mu}$



$\Rightarrow dr = \frac{-\sqrt{3}}{2\mu^2}\frac{du}{\cos r}$

In ΔOPQ,

$\frac{y}{\sin r} = \frac{3}{\sin(150 - r)}$ …(i)

Differentiate (i) and put r = 30° for $\mu = \sqrt{3}$

$\therefore\ |dy| = \frac{2}{3}d\mu = 0.2133$

15. 00001.33

Sol. For K_α; $\sqrt{\nu} = \sqrt{\frac{3R}{4}}(z - b)$

$K_\beta;\ \sqrt{\nu} = \sqrt{\frac{8R}{9}}(z-b)$

$\frac{K_\beta}{K_\alpha} = \frac{\sqrt{\frac{8}{9}}}{\sqrt{\frac{3}{4}}} = \left(\frac{4}{3}\right)^{3/2}\frac{1}{\sqrt{2}} \Rightarrow n = 1.33$

16. 00016.67

Sol. $\vec{F} = q(\vec{v}\times\vec{B})$

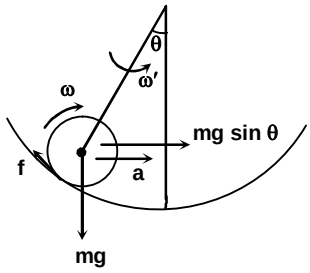
$(3\hat{i}+4\hat{j})\times10^{-10} = 2\times10^{-9}(x\hat{i}+y\hat{j})\times3\times10^{-3}\hat{k}$

Solve: $x = -\frac{400}{6};\ y = \frac{300}{6} \Rightarrow |x+y| = \frac{100}{6} = 16.67$

17. 00001.76

Sol. $mg\sin\theta - f = ma$ …(i)

$f_r = \frac{2}{5}mr^2\alpha$ …(ii)



$\Rightarrow a = \frac{5}{7}g\sin\theta$

For small θ; $a = \frac{5}{7}g\theta = \left(\frac{5g}{7(R-r)}\right)x$

$\Rightarrow T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$

18. 00312.50

Sol. $B = \frac{dP}{dV/V} \Rightarrow \frac{dV}{V} = \frac{dP}{B} \Rightarrow dR = \frac{R}{3}\left(\frac{dP}{B}\right)$

$dR = \frac{3}{100\times3}\times\frac{50}{8\times10^{-3}}\times\frac{1}{2\times10^{9}} \simeq 312.5\ \text{Å}$

Chemistry

PART – II

SECTION – A

19. B

Sol. $pH = \frac{PKa_1 + PKa_2}{2}$

$= \frac{2.4 + (14 - 4.6)}{2}$

= 5.9

20. A, B, D

21. A, B, C, D

Sol. Properties of Baryllium.

22. A, B, D

Sol. $P_4 + 3NaOH + 3H_2O \longrightarrow PH_3 + 2NaH_2PO_2$

$(White)$

23. A

24. B, C

Sol. $(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$

$2CrO_3 \xrightarrow{\Delta} Cr_2O_3 + \frac{3}{2}O_2$

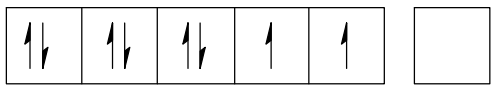
$2K_2Cr_2O_7 \xrightarrow{\Delta} 2K_2CrO_4 + Cr_2O_3 + \frac{3}{2}O_2$

$Al_2O_3 + Cr \longrightarrow$ No reaction

25. B

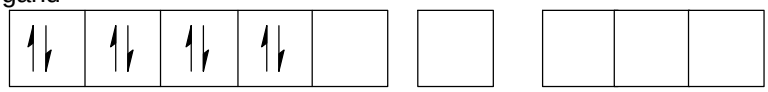
Sol. (P) $[Ni(CN)_4]^{2-}$

O. S of Ni is +2

$Ni^{+2} \Rightarrow 3d^8 \Rightarrow$  Paramagnetic

3d 4s

In presence of ligand

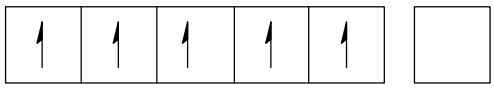
$Ni^{+2} \Rightarrow 3d^8 \Rightarrow$  Diamagnetic

3d 4s 4p

dsp^2

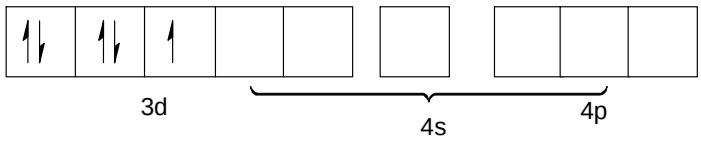
(Q) $K_3[Fe(CN)_6]$

Oxidation state of Fe = +3

$Fe^{+3} \Rightarrow 3d^5 4s^0 \Rightarrow$  Paramagnetic

3d 4s

In presence of ligand



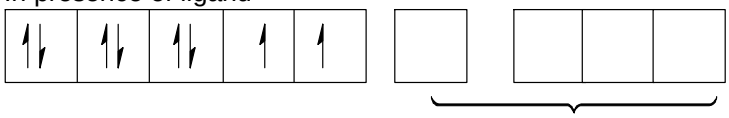
Paramagnetic low spin

d^2sp^3

(R) $Ni^{2+} \Rightarrow$



In presence of ligand

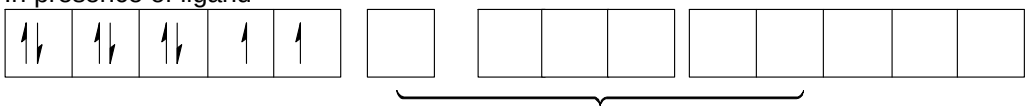


Paramagnetic

sp^3

(S) $[Ni(NH_3)_6]SO_4$

In presence of ligand



sp^3d^2 paramagnetic

26. A

Sol. Ag^+ - $Ag_2S \downarrow$
- AgI ↓ yellow
- 1st group

Zn^{2+} - 4th group
- ZnS ↓
- Soluble in KI
- form Amphoteric Oxide

Pb^{2+} - Pbs ↓
- PbI_2 ↓ yellow
- 1st and 2nd group
- form Amphoteric oxides

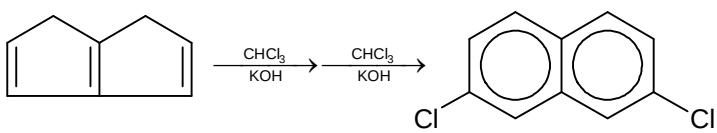
Bi^{3+} - Bi_2S_3 ↓
- BiI_3 ↓ Black
- BiO^+X^- (white ppt on dilution with water)
- 2nd group

27. D

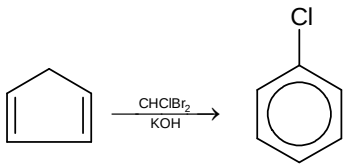
Sol. P – Chiral, G. I., 2 chiral centers
Q – Chiral, G.I., 2 chiral centers
R – Chiral, G. I., 3 chiral centers
S – Achiral, G. I., plane of symmetry

28. A

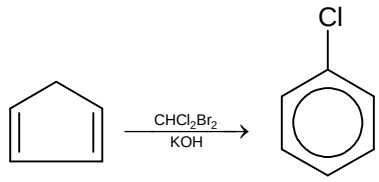
Sol. (P)



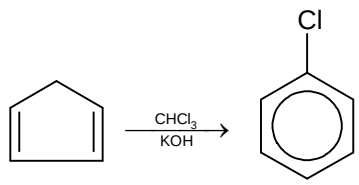
(Q)



(R)



(S)



SECTION – D

29. 00042.22

Sol. For n mole

$\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT$ for negligible volume $b \approx 0$

$PV^2 - nRTV + n^2a = 0$

Quadratic in V thus,

$V=\frac{nRT \pm \sqrt{n^2R^2T^2-4an^2P}}{2}$

V has one value at given pressure and temperature thus discriminant = 0

$n^2R^2T^2 = 4a\,n^2P$

P = 42.22 atm.

30. 00046.06

Sol. Meq of H_2SO_4 = Meq of NaOH

N of $H_2SO_4 = \frac{84.6}{5}$

Weight of H_2SO_4 in 1 lit = $\frac{84.6}{5}\times 49$

Weight of H_2SO_4 in 1 ml = $\frac{84.6\times 49}{5\times 1000}$

Purity = $\frac{84.6\times 49\times 100}{5\times 1.8\times 1000}=46.06\%$

31. 00008.50

Sol. $u=\frac{h}{m\lambda}=\frac{6.6\times 10^{-34}}{1.6\times 10^{-27}\times 0.01\times 10^{-9}}=4.125\times 10^4\,m/s$

$QV = 1/2\ mu^2$

Putting all values we get V

V = 8.50 volt.

32. 00003.82

Sol. Equivalent of oxygen = Equivalent of metal

$\frac{20}{8}=\frac{80}{E}$

E = 32

N.f = $\frac{64}{32}=2$

2F charge is required for 64 gm metal

1F charge is required for 32 gm metal

$\frac{60\times60\times3.2}{96500}$ Coulomb charge is required for $\frac{32\times60\times60\times3.2}{96500}$ gm metal

= 3.82 gm

33. 00062.10

Sol.

$$A(g)\longrightarrow B(g)+C(g)$$

At time $t=0$	p	0	0
At time $t=20$ min	$p-x$	x	x
At time $t=\infty$	0	p	p

$p + x = 300$

$2p = 500$

$p = 250$

$x = 50$

$K=\frac{2.303}{20}\log1.25$

$t_{1/2}=\frac{0.693}{K}=62.10$ min

34. 00001.43

Sol. $HCl + NH_4OH \rightarrow NH_4Cl + H_2O \quad \Delta H = -12.27$

$H^+ + Cl^- + NH_4OH \rightarrow NH_4^+ + Cl^- + H_2O \quad \Delta H = -12.27$ …1

$NH_4OH \rightarrow NH_4^+ + OH^- \quad \Delta H = x$ … 2

From equation (1) and (2)

$H^+ + OH^- \rightarrow H_2O \quad \Delta H = -12.27 - x$

We know that

$-12.27 - x = -13.7$

$X = 1.43$ Kcal

35. 00073.75

Sol. $PCl_5 \rightleftharpoons PCl_5 + Cl_2$

1	0	0
$1-\alpha$	α	α

$\frac{M_{eq}}{M_{initial}}=\frac{1}{1+\alpha}$

$\frac{208.5}{120}=1+\alpha$

$\alpha = 0.7375$

Or 73.75%

36. 00344.80

Sol. Work is done against constant pressure and thus irreversible.

$\Delta V = 4$ lit

$P = 1$ atm

$w = -1 \times 4$ lit-atm $= - 1 \times 4 \times 101.3 = - 405.2$ Joule

$\Rightarrow \Delta E = a + w$

$\therefore \Delta E = 750 - 405.2$

$\Delta E = +344.80$ J

Mathematics PART – III

SECTION – A

37. A, D

Sol. $f'(x) = 3(x - 1)(x + 1)$

$\Rightarrow$ f is strictly non-decreasing in $(-\infty, -1]$ and $[1, \infty)$ and non-increasing in $[-1, 1]$

$f(-1) = 3, f(1) = -1, f(2) = 3$

$f(x) = 0$ has three distinct roots x_1, x_2, x_3 i.e. $-2 < x_1 < -1 < x_2 < 1 < x_3 < 2$

So $f(x) = x_1$ has one solution

$f(x) = x_2$ = has 3 solution

$f(x) = x_3$ has 3 solution

38. A, C

Sol. $\Delta = 0$

$$\Rightarrow \begin{vmatrix} \cos\theta & \sin\theta & \cos\theta \\ \sin\theta & \cos\theta & \sin\theta \\ \cos\theta & \sin\theta & -\cos\theta \end{vmatrix} = 0$$

$\Rightarrow \cos\theta \cdot \cos 2\theta = 0$

$\therefore \ \theta = \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}$

But $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ (rejected)

Because line are parallel, so system is inconsistent

39. A, C

Sol. (i) We have $u = \frac{4}{4-x^2} + \frac{9x^2}{9x^2-1} = 1 + \frac{35x^2}{-9x^4+37x^2-4} = 1 + \frac{35}{37 - \left(\left(3x - \frac{2}{x}\right)^2 + 12\right)}$

Since $x \in \left(-2, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 2\right)$. So u reaches minimum value $\frac{12}{5}$, when $x = \pm\sqrt{\frac{2}{3}}$

(ii) Using A.M. and G.M.

$$u \geq 2\sqrt{\frac{4}{4-x^2} \cdot \frac{9}{9-y^2}}$$

$$\Rightarrow \frac{12}{\sqrt{36 - 9x^2 - 4y^2 + (xy)^2}} = \frac{12}{\sqrt{37 - 9x^2 - 4y^2}} \geq \frac{12}{\sqrt{37 - 2\sqrt{36(xy)^2}}} = \frac{12}{5}$$

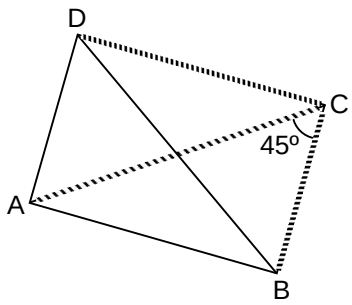
40. A, B

Sol. $\frac{1}{3}AD\left(\frac{1}{2}BC \cdot AC \cdot \sin 45^o\right) \geq V_{ABCD} = \frac{1}{6}$

So, $AD \cdot BC \cdot \frac{AC}{\sqrt{2}} \geq 1$

$$3 = AD + BC + \frac{AC}{\sqrt{2}} \geq 3\sqrt{AD \cdot BC \cdot \frac{AC}{\sqrt{2}}} \geq 3$$

Equality holds if and only if $AD = BC = \frac{AC}{\sqrt{2}} = 1$



So AD is perpendicular to the face ABC. $DC = \sqrt{AD^2 + AC^2} = \sqrt{3}$

41. A, B, C, D

Sol. $(2x + 1)^2 = 1996(1997)^n + 1997$ for n = 1, we get $(2x + 1)^2 = 1997^2$

So $2x + 1 = \pm 1997 \Rightarrow x = 998, -999$

42. A, B

Sol. $T_r = \sum_{r=1}^{n} \cot^{-1}\left(r^2 + \frac{3}{4}\right)$

$$T_r = \tan^{-1}\left(\frac{4}{4r^2+3}\right) = \tan^{-1}\left(\frac{1}{r^2+\frac{3}{4}}\right)$$

$$T_r = \tan^{-1}\left(\frac{1}{1+\left(r^2-\frac{1}{4}\right)}\right)$$

$$T_r = \tan^{-1}\left(r+\frac{1}{2}\right) - \tan^{-1}\left(r-\frac{1}{2}\right)$$

$$S_n = \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$$

$$\lim_{n\to\infty} S_n = \tan^{-1}\infty - \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2} - \tan^{-1}\frac{1}{2} = \tan^{-1} 2$$

43. C

Sol. $e^{-x} + e^{-x^3} = x$

Let $f(x) = e^{-x} + e^{-x^3} \Rightarrow f'(x) = -e^{-x} - 3x^2 \cdot e^{-x^3} < 0$

$f(-\infty) = \infty$, $f(\infty) = 0 \Rightarrow f(x) = x$ has one solution

$\sin^5 x \le \sin^2 x$; $\cos^3 x \le \cos^2 x$

$\Rightarrow \sin^5 x + \cos^3 x \le \sin^2 x + \cos^2 x \Rightarrow \sin^5 x + \cos^3 x \le 1$

Equality holds if $\sin^5 x = \sin^2 x$ and $\cos^3 x = \cos^2 x$

$\Rightarrow \sin x = 0, \cos x = 1$ or $\sin x = 1, \cos x = 0$

$\Rightarrow x = \frac{\pi}{2}, 2\pi$

44. A

Sol. $f(x) = \frac{1-\sqrt{17}}{2x} \Rightarrow f'(x) = \frac{\sqrt{17}-1}{2x^2}$

45. B

Sol. Length of common chord = $2\sqrt{9 - \frac{9}{4}} = 3\sqrt{3}$

Angle between circles = $\pi - \cos^{-1}\left(\frac{9+9-9}{2\cdot 3\cdot 3}\right) = \frac{2\pi}{3}$

46. B

Sol. Let $I_n = \int_0^{\pi/2} \sin^n x\,dx \Rightarrow I_n = \frac{n-1}{n} \cdot I_{n-2}$

Let $P_n = \int_0^{\pi/4} \tan^n x\,dx \Rightarrow P_n + P_{n-2} = \frac{1}{n-1}$

Let $I_n = \int_0^1 x(1-x)^n\,dx \Rightarrow I_n = \frac{1}{n+1} - \frac{1}{n+2}$

SECTION – D

47. 00002.00

Sol. $g(\alpha) = g'(\alpha) = g''(\alpha) = 0 \Rightarrow g(x) = a(x-\alpha)^3(x-\beta)$ and $\lim_{x\to\alpha} \frac{f(x)}{g(x)} = 0$

$\Rightarrow f(x) = b(x-\alpha)^4$

$h(x) = f(x)g'(x) + g(x)f'(x) = \frac{d}{dx}\left(f(x)g(x)\right)$

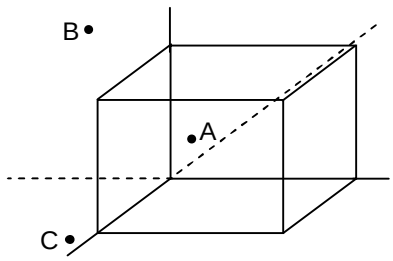
$= \frac{d}{dx}\left(ab(x-\alpha)^7(x-\beta)\right) = ab((x-\alpha)^7 + 7(x-\beta)(x-\alpha)^6)$

$= ab(x-\alpha)^6(x-\alpha+7x-7\beta) = ab(x-\alpha)^6(8x-\alpha-7\beta)$

$\therefore$ h(x) has two different real solutions

48. 00002.00

Sol. A(3, 1, 0) inside cube
B(1, 1, 1) outside cube
C(2, 1, 3) outside cube
Number of intersection point is 2



49. 00006.00

Sol. $|A| = -18$

$A\,adj(A) = -18\,I$

$$\Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix} = \begin{bmatrix} -18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & -18 \end{bmatrix}$$

$\Rightarrow -3\sum a_{ij} = -3(18) \Rightarrow \frac{1}{3}\sum a_{ij} = 6$

50. 00007.00

Sol. $\frac{\sin 1^\circ \sin 2^\circ \sin 3^\circ \ldots\ldots \sin 179^\circ}{\sin 2^\circ \sin 4^\circ \ldots\ldots \sin 178^\circ} = \sin 1^\circ \sin 3^\circ \sin 5^\circ \ldots\ldots \sin 179^\circ$

$\frac{\left(\frac{1}{2^{44}} \times \frac{1}{\sqrt{2}} \sin 2^\circ \sin 4^\circ \ldots\ldots \sin 88^\circ\right)^2}{(\sin 2^\circ \sin 4^\circ \ldots\ldots \sin 88^\circ)^2} = \frac{1}{2^{89}} \Rightarrow \lambda = 89$

51. 00036.00

Sol. $C_1 : \dfrac{(x-3)^2}{9} + \dfrac{(y-2)^2}{4} = 1$

$\therefore$ $A_1 = (6, 2)$, $A_2 = (0, 2)$

$\therefore$ $|PA_1 - PA_2| = 3\sqrt{2}$

Clearly locus of P is hyperbola for which

$A_1A_2 = 2ae = 6$ and $2a = 3\sqrt{2} \Rightarrow e = \sqrt{2}$

$\therefore$ Locus of P is rectangular hyperbola

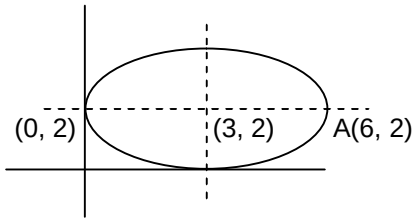
Equation of conic $C_2 : (x-3)^2 - (y-2)^2 = \dfrac{9}{2}$

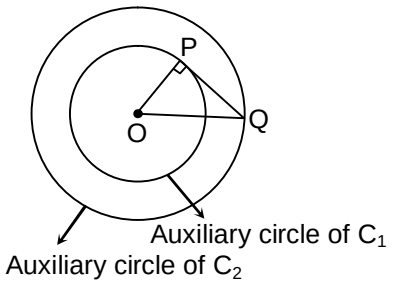
Now $D_1 = 2ae = 6$

$D_2 = b^2 = \dfrac{9}{2}$

$D_3 : L = \sqrt{9 - \dfrac{9}{2}} = \dfrac{3}{\sqrt{2}}$

$\therefore \left(\dfrac{D_1D_2}{D_3^2}\right)^2 = 36$





52. 00002.00

Sol. $\left(y + \dfrac{2}{3}\right)^2 = 2\left(x - \dfrac{10}{9}\right)$

Clearly for three distinct normal x-coordinate > 2a

$\Rightarrow x - \dfrac{10}{9} > 2 \cdot \dfrac{1}{2} \Rightarrow h > \dfrac{19}{9} \Rightarrow [h]_{least} = 2$

53. 00003.00

Sol. $x = 6q + r$ where $0 \le r \le 5$

$\therefore \left[\dfrac{x}{2}\right] - \left[\dfrac{x}{3}\right] = \dfrac{x}{7} \Rightarrow \left[\dfrac{r}{2}\right] - \left[\dfrac{r}{3}\right] = \dfrac{-9}{7} + \dfrac{r}{7}$

for r = 0, we get $q = 0 \Rightarrow x = 0$

for r = 1, we get $q = 1 \Rightarrow x = 7$

for r = 2, we get $q = -5 \Rightarrow x = -28$

for r = 3, we get $q = 3 \Rightarrow x = 21$

for r = 4, we get $q = -3 \Rightarrow x = -14$

for x = 5, we get $q = -2 \Rightarrow x = -7$

$\therefore$ N = sum of all value = $-21 \Rightarrow \left|\dfrac{N}{7}\right| = 3$

or Put $x = 7k$ where $k \in I$

54. 00005.00

Sol. $f(x) = \dfrac{\sin x}{x}$ for $x \in \left(0, \dfrac{\pi}{2}\right)$; $f'(x) = \dfrac{x\cos x - \sin x}{x^2} = \cos x\left(\dfrac{x - \tan x}{x^2}\right) = -ve$ for $x \in \left(0, \dfrac{\pi}{2}\right)$

$\Rightarrow$ f(x) is decreasing function for $x \in \left(0, \dfrac{\pi}{2}\right)$

$f''(x) = \left(\dfrac{-x^2\sin x - 2x\cos x + 2\sin x}{x^3}\right)$

Let $h(x) = -x^2 \sin x - 2x \cos x + 2 \sin x$

$h'(x) = -x^2 \cos x = -ve$ for $x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow$ h(x) is decreasing function for $x \in \left(0, \frac{\pi}{2}\right)$; $h(x) < h(0)$ for $x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow -x^2 \sin x - 2x \cos x + 2 \sin x < 0 \ \forall \ x \in \left(0, \frac{\pi}{2}\right)$

$\Rightarrow f''(x) < 0 \ \forall \ x \in \left(0, \frac{\pi}{2}\right) \Rightarrow$ f(x) is concave down on $\left(0, \frac{\pi}{2}\right)$

$\therefore$ for any $x_1, x_2, x_3 \in \left(0, \frac{\pi}{2}\right)$

We can write $f\left(\frac{x_1 + x_2 + x_3}{3}\right) \geq \frac{f(x_1) + f(x_2) + f(x_3)}{3}$

$x_1 = A.\ x_2 = B,\ x_3 = C$

$$\frac{\sin\left(\frac{A+B+C}{3}\right)}{\frac{A+B+C}{3}} \geq \frac{\frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C}}{3}$$

$$\frac{9}{\pi}\sin\frac{\pi}{3} \geq \frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C}, \quad M = \frac{9\sqrt{3}}{2\pi} \Rightarrow \left[\frac{2M\pi}{3}\right] = \left[3\sqrt{3}\right] = 5$$